

Photosynthesis

Learning Booklet

Name: _____ Class: _____ Date: _____

□ What you are learning

- State the reactants and products of photosynthesis.
- Recall photosynthesis is endothermic and write the equations.
- Describe the uses of the glucose a plant makes.
- Explain how light, carbon dioxide and temperature change the rate (limiting factors).
- Interpret a rate graph and describe the required practical with pondweed.

Common mistakes (read first)

- Swapping the gases. The plant takes IN carbon dioxide and gives OUT oxygen — for example, the bubbles from pondweed are oxygen.
- Calling light a raw material. The raw materials (reactants) are water and carbon dioxide; light is the energy source.
- Saying it happens 'in the leaf'. Be exact: it happens in the chloroplast inside the leaf cell.
- Forgetting plants respire too. Plants photosynthesise AND respire — respiration never stops, day or night.
- Writing the equation backwards. Reactants (CO_2 + water) go on the LEFT; products (glucose + oxygen) on the RIGHT.

Key vocabulary

Photosynthesis	how a plant makes its own food (glucose) using light energy
Reactants	what you start with: carbon dioxide and water
Products	what is made: glucose and oxygen
Endothermic	a reaction that takes IN energy from the surroundings
Chlorophyll	the green pigment that absorbs (traps) light energy
Chloroplast	the part of a plant cell where photosynthesis happens
Limiting factor	the thing in shortest supply that slows the rate down
Rate	how fast the reaction happens (e.g. bubbles per minute)

How photosynthesis works

Plants cannot move to find food, so they make their own using light energy from the Sun. Photosynthesis is endothermic, which means it takes in energy.

Light energy is trapped by chlorophyll inside the chloroplasts. The energy joins carbon dioxide (from the air) and water (from the soil) together.

This makes glucose, a sugar the plant uses for energy and growth. Oxygen is made as well and is released into the air.

Word & symbol equation

carbon dioxide + water $\xrightarrow{\text{(light + chlorophyll)}}$ glucose + oxygen



Diagram 1 — inputs & outputs

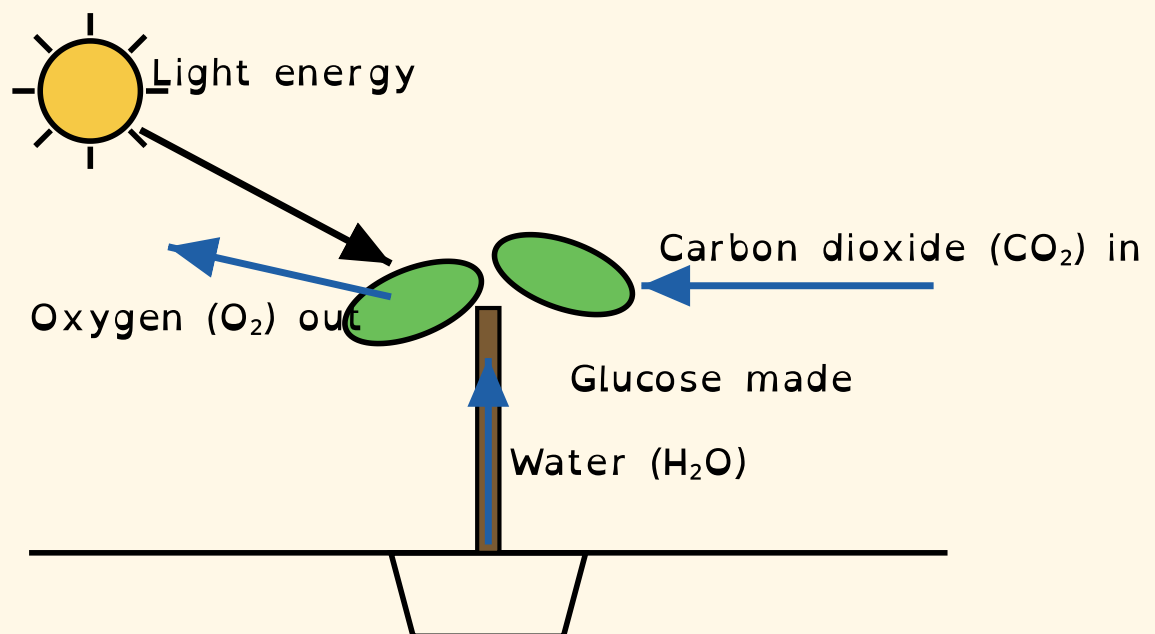


Diagram 1: What goes into and comes out of a plant

Remember (content)

- IN (reactants): carbon dioxide + water. OUT (products): glucose + oxygen.
- Where: chloroplasts in the leaf. Energy: light, trapped by chlorophyll.

- Photosynthesis is endothermic (takes IN energy). Plants also respire — day and night.

Diagram 2 — inside the leaf cell

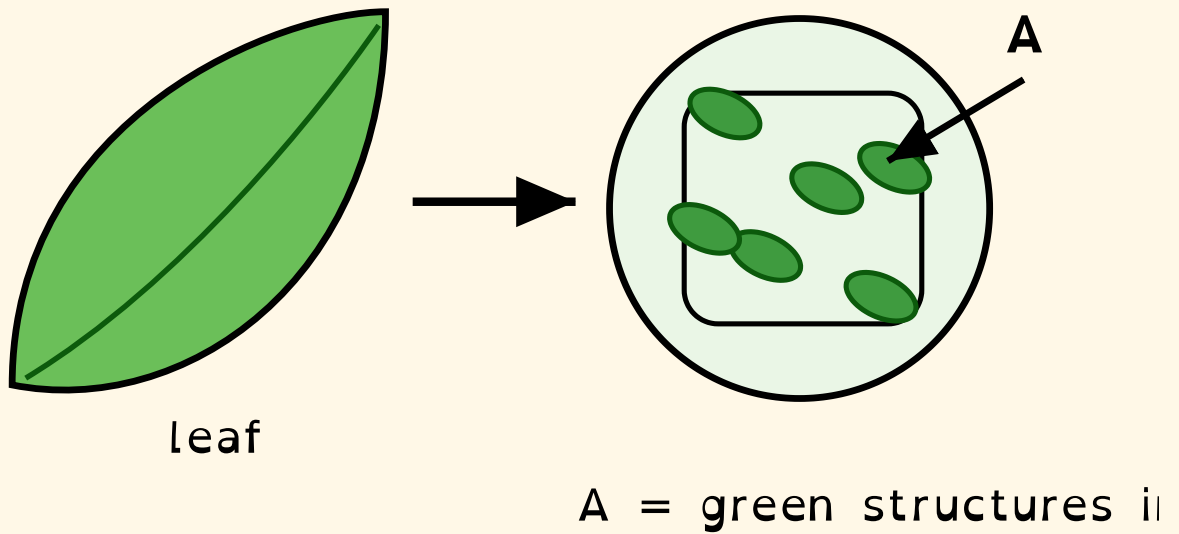


Diagram 2: Light is trapped by chlorophyll in the chloroplast (A)

What the glucose is used for

- Used in respiration to release energy.
- Used for growth (making new cells).
- Stored as starch for later.
- Used to make cellulose for strong cell walls.
- Used (with nitrate ions) to make amino acids for proteins.

What changes the rate? (limiting factors)

- Light intensity — more light means a faster rate, until something else runs short.
- Carbon dioxide concentration — more CO₂ means a faster rate, up to a point.
- Temperature — warmth speeds it up, but too hot damages the enzymes and it slows down.
- Amount of chlorophyll — less chlorophyll (e.g. diseased leaves) means a slower rate.

Diagram 3 — rate vs light intensity

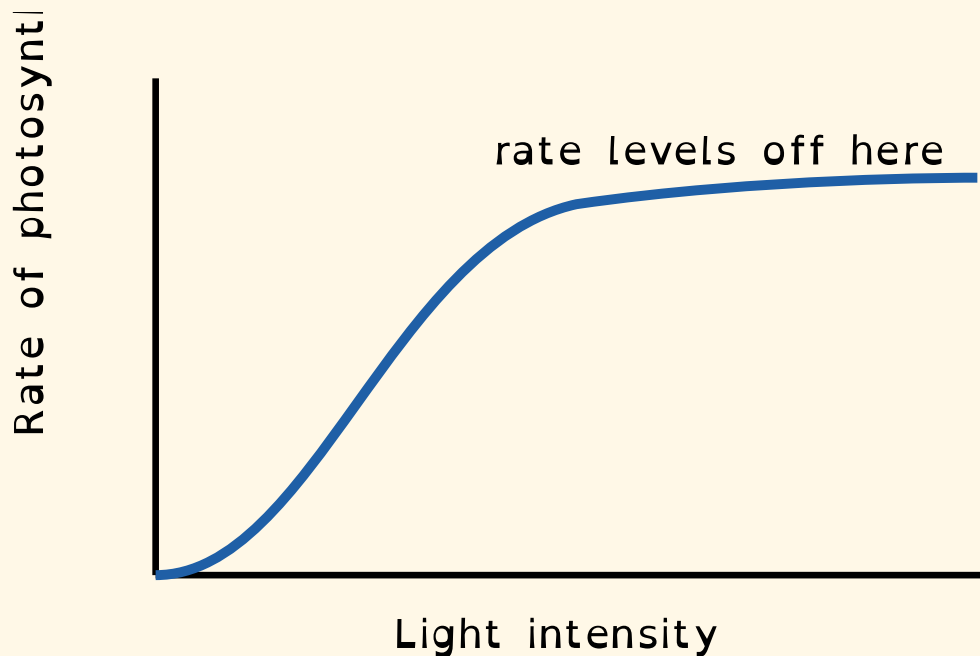


Diagram 3: Rate of photosynthesis vs light intensity (it levels off)

□ How to answer an 'explain' question

- 1 Say what happens (the fact).
- 2 Say WHY, using a key word (light / chlorophyll / chloroplast / limiting factor).
- 3 Link back to the question (so... / this means...).
- Example: 'There is no light, so chlorophyll cannot trap energy, so no glucose is made.'

Questions — they get harder as you go

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Q1 [1 mark]

Recall ● ○ ○ ○ ○

State the two reactants (raw materials) needed for photosynthesis.

Q2 [1 mark]

Recall ● ○ ○ ○ ○

Name the two products made in photosynthesis.

Q3 [2 marks]

Knowledge ● ● ○ ○ ○

Write the word equation for photosynthesis.

Q4 [1 mark]

Knowledge ● ● ○ ○ ○

Photosynthesis is an endothermic reaction. What does 'endothermic' mean?

□ **Hint:** Think about energy — is it taken in or given out?

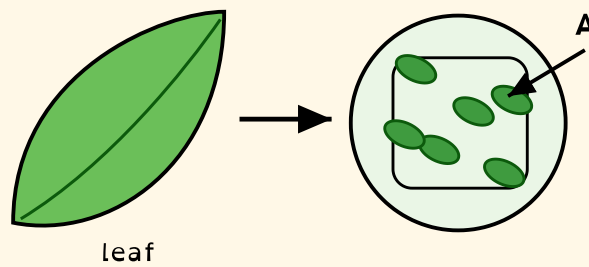
Q5 [2 marks]

Diagram ● ● ● ○ ○

Look at the diagram below.

(a) Name structure A.

(b) Name the green pigment that structure A contains.



A = green structures inside the leaf cell

Q6 [3 marks]

Apply ● ● ● ○ ○

Give three different ways a plant uses the glucose it makes.

□ **Hint:** Think: energy (respiration), growth, and storage.

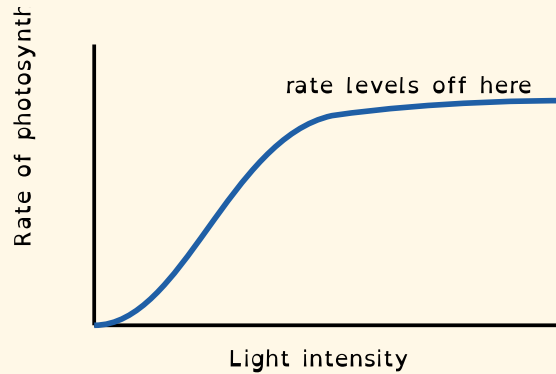
Q7 [4 marks]

Graph / Explain ●●●●○

Look at the graph below.

(a) Describe what happens to the rate of photosynthesis as light intensity increases.

(b) Explain why the rate levels off (stops increasing).



Sentence starters:

(a) As light intensity increases, the rate ____.

(b) The rate levels off because ____ is now the limiting factor.

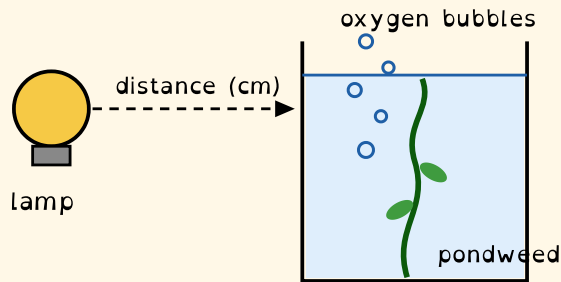
Hint: When light is no longer in short supply, another factor (carbon dioxide or temperature) becomes the limiting factor.

Q8 [3 marks]

Required practical ●●●●○

A student investigates the effect of light intensity on photosynthesis. They move a lamp closer to pondweed and count the gas bubbles each minute (see diagram).

- (a) Name the gas in the bubbles.
- (b) Give one variable they must keep the same (control).
- (c) State one way they could make the results more accurate.



□ **Hint:** Control variables here include temperature and the amount of carbon dioxide. For accuracy, repeat and find a mean.

Q9 [6 marks]

Exam-style (problem solving) ● ● ● ● ●

A farmer grows tomatoes in a greenhouse during winter. The tomatoes are growing slowly. Explain how the farmer could increase the rate of photosynthesis so the tomatoes grow faster. In your answer, refer to more than one limiting factor.

Sentence starters:

One way is to ____ because ____.

Another way is to ____ because ____.

The farmer must also think about ____ (e.g. cost).

□ **Hint:** Use the three limiting factors: light, carbon dioxide and temperature. Say **WHAT** to change and **WHY** it speeds up the rate.

Self-reflection

How confident do you feel with photosynthesis now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

One key word I want to remember:

The question I found hardest was ____ because:

One thing I will revise before the next lesson:

Teacher assessment

Success criteria (tick if met) — total: 22 marks

- ☐ States reactants and products correctly (Q1–Q2)
- ☐ Writes the word equation; explains 'endothermic' (Q3–Q4)
- ☐ Identifies the chloroplast/chlorophyll from the diagram (Q5)
- ☐ Gives three uses of glucose (Q6)
- ☐ Interprets the rate graph and explains the plateau (Q7)
- ☐ Describes the required practical: gas, control variable, accuracy (Q8)
- ☐ Q9 (6 marks): applies more than one limiting factor to the greenhouse problem

Red — needs re-teaching

Amber — nearly there

Green — secure

What went well (WWW):

Even better if (EBI):

Next step / target:

Teacher answer key

1. Q1: Carbon dioxide and water.
2. Q2: Glucose and oxygen.
3. Q3: carbon dioxide + water $\rightarrow$ glucose + oxygen (light + chlorophyll above the arrow).
4. Q4: It takes in energy from the surroundings (here, light energy).
5. Q5: (a) Chloroplast. (b) Chlorophyll.
6. Q6: Any three of: respiration (energy); growth; stored as starch; making cellulose for cell walls; making amino acids/proteins (with nitrate ions).
7. Q7: (a) The rate increases as light intensity increases. (b) It levels off because another factor (carbon dioxide concentration or temperature) becomes the limiting factor — light is no longer limiting.
8. Q8: (a) Oxygen. (b) Any one: temperature / amount of carbon dioxide / same piece of pondweed. (c) Repeat and calculate a mean (or count for a fixed time).
9. Q9 (6 marks, indicative): Increase light (lamps) — gives more energy for the reaction; add carbon dioxide (e.g. CO₂ burner) — more raw material; raise the temperature (heater) — enzymes work faster (but not too hot or enzymes denature). Must reference more than one factor and link change $\rightarrow$ faster rate; may mention balancing cost.